



Medical Microbiology

Lecture Fifteen

Parasitology

Third stage- College of Dentistry

Dr. Dheyaa Naji Hamza

Dheyaa.naji@alkafeel.edu.iq

Introduction:

- **Parasitology:** is the study of parasites, their life cycles, interactions with hosts, and their impact on human health.
- Parasites cause significant morbidity and mortality worldwide.
- Many parasitic infections are linked to poor sanitation and contaminated water.

Parasites are classified into three main groups:

1-Protozoa: Microscopic, single-celled organisms.

Examples: Giardia lamblia, Entamoeba histolytica.

2-Helminthes: Multicellular parasitic worms.

Examples: Ascaris lumbricoides, Schistosoma species.

3-Arthropods: Invertebrates such as insects and arachnids that may act as parasites or vectors of disease.

Examples: ticks and lice.

Protozoan Parasites – General Characteristics:

- 1-Unicellular eukaryotic organisms.
- 2-Reproduce through **binary fission**.
- 3-Transmitted via fecal-oral route, vector by bites, or direct contact.
- 4-Exist in two forms:
 - **Trophozoite**: Active and motile stage.
 - **Cyst**: Dormant, resistant for survival outside the host.



Giardia lamblia (also called **Giardia intestinalis** or **Giardia duodenalis**):

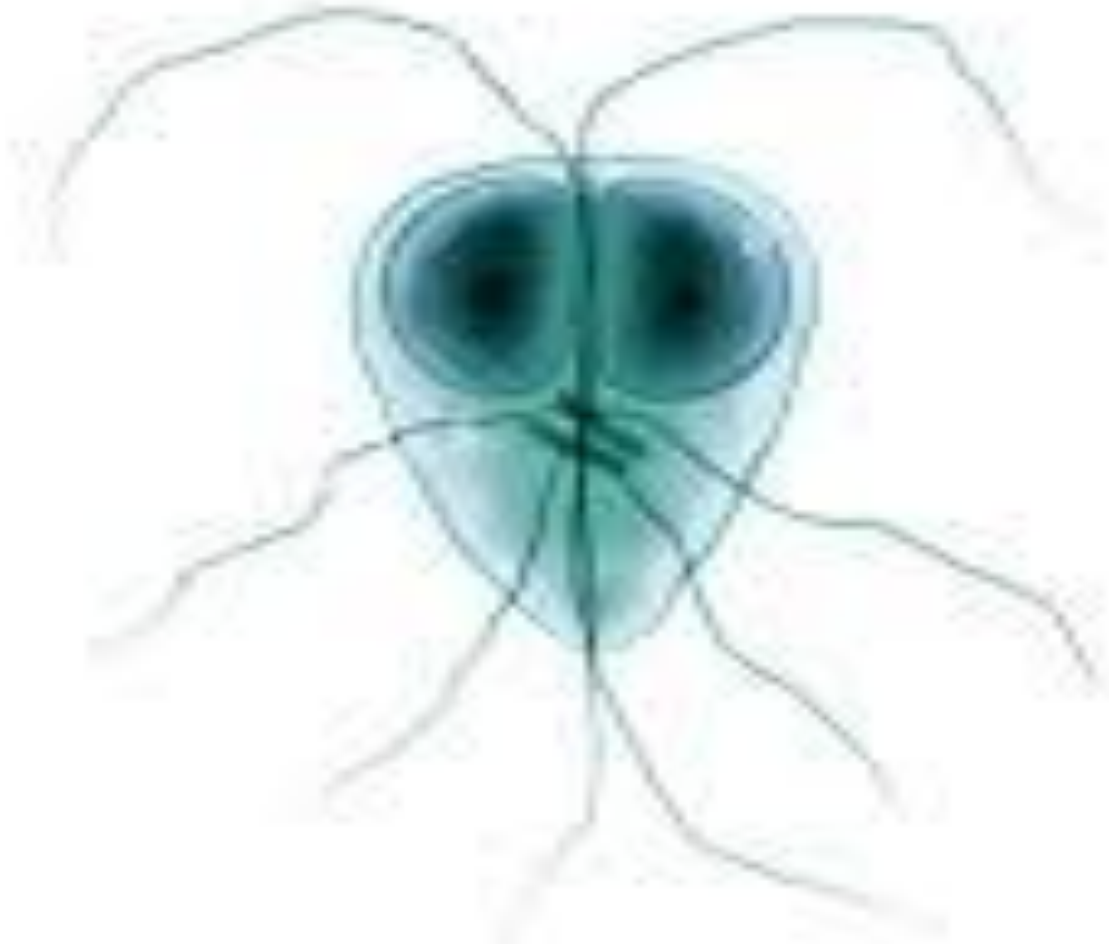
- is a protozoan parasite that causes **giardiasis**, a common intestinal infection.
- It is a significant cause of diarrheal disease worldwide, especially in areas with poor sanitation.

It has two morphological forms:

1- Trophozoite: The active, motile, feeding form that inhabits the small intestine.

2- Cyst: The infective stage, resistant and dormant form responsible for transmission.

GIARDIA INTESTINALIS



Trophozoite



Cyst

Life Cycle of Giardia lamblia:

1-Infection: Infection begins when mature cysts are ingested through contaminated water, food, or unwashed hands.

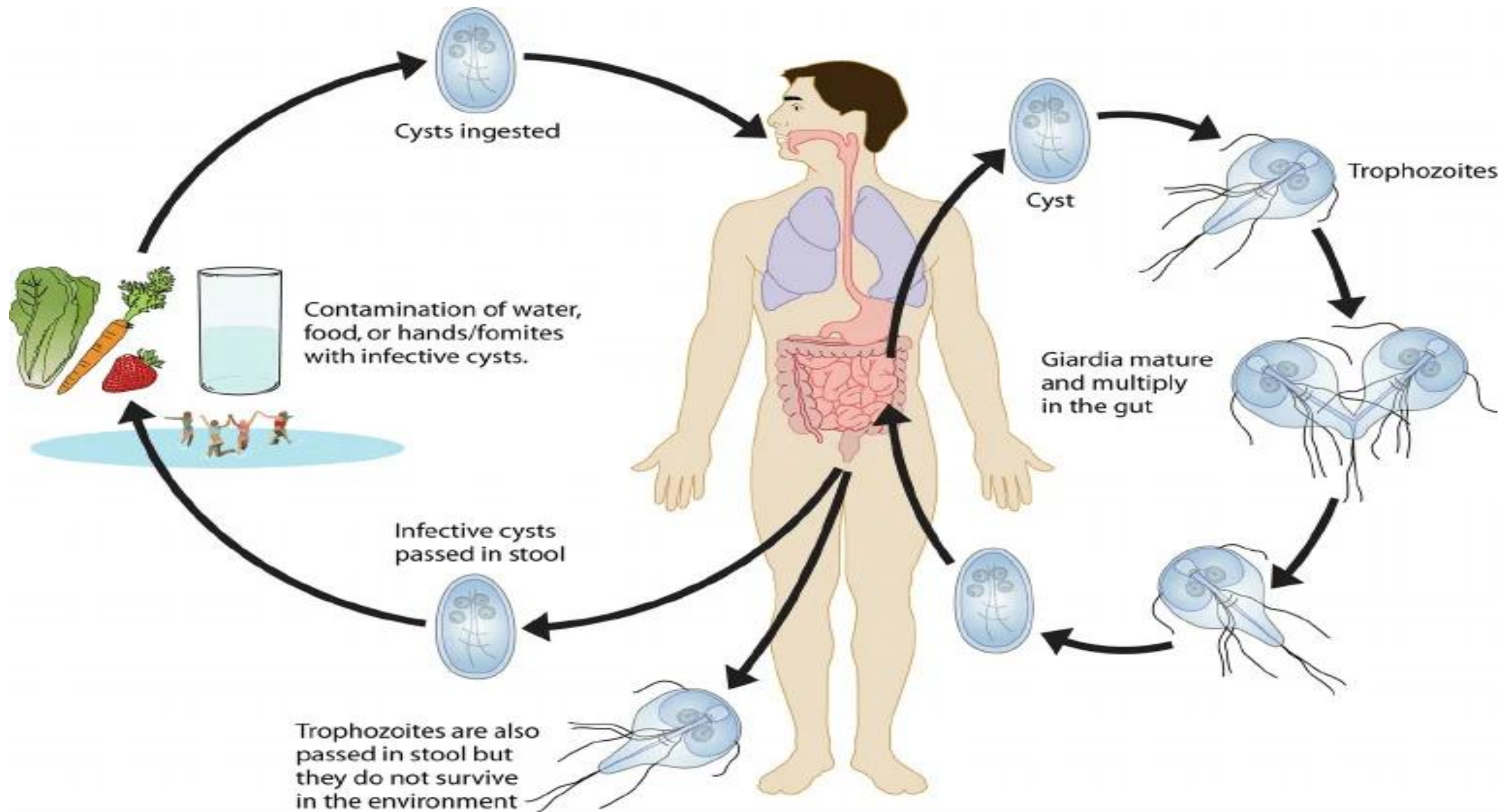
2-Excystation: Occurs in the **small intestine**, releasing active trophozoites.

3-Multiplication: Trophozoites **multiply by binary fission in the small intestine**.

4-Attachment: Trophozoites adhere to the intestinal mucosa using a **ventral adhesive disc** without invading the tissue.

5-Encystation: As parasites move toward the **colon**, they form new cysts.

6-Transmission: Cysts are passed in feces and can survive in the environment, infecting new hosts.



Transmission:

- 1-Contaminated water (most common; drinking water).
- 2-Food contaminated by infected individuals.
- 3-Person-to-person contact.
- 4-Zoonotic transmission from animals to humans.

Pathogenesis:

- 1-Giardia **attaches** to the **small intestine** and disrupts nutrient absorption.
- 2-Causes **malabsorption** and **watery diarrhea**.

Clinical Features:

- 1-Watery, foul-smelling diarrhea.
- 2-Abdominal cramps, bloating, and weight loss.
- 3-Symptoms can be **chronic in immunocompromised patients.**

Diagnosis:

- 1-Stool examination: Detection of cysts or trophozoites using microscopy.
- 2-Stool antigen test (ELISA or DFA): more sensitive.
- 3-PCR test: highly specific but expensive.

Treatment:

- 1- **Metronidazole** is the first-line treatment.
- 2- **Tinidazole** (alternative, single-dose treatment).

Preventive measures:

- 1- Drinking filtered or boiled water in endemic areas.
- 2- Proper hand hygiene and sanitation.
- 3- Avoiding contaminated food and water sources.
- 4- Treating infected individuals to prevent spread.

Entamoeba histolytica and Amebiasis:

Entamoeba histolytica: is a protozoan parasite that causes **amebiasis**, a disease affecting the intestines and, in severe cases, other organs like the liver.

It has two morphological forms:

1-Trophozoite (active form):

Amoeboid in shape, **single nucleus**, with central karyosome.

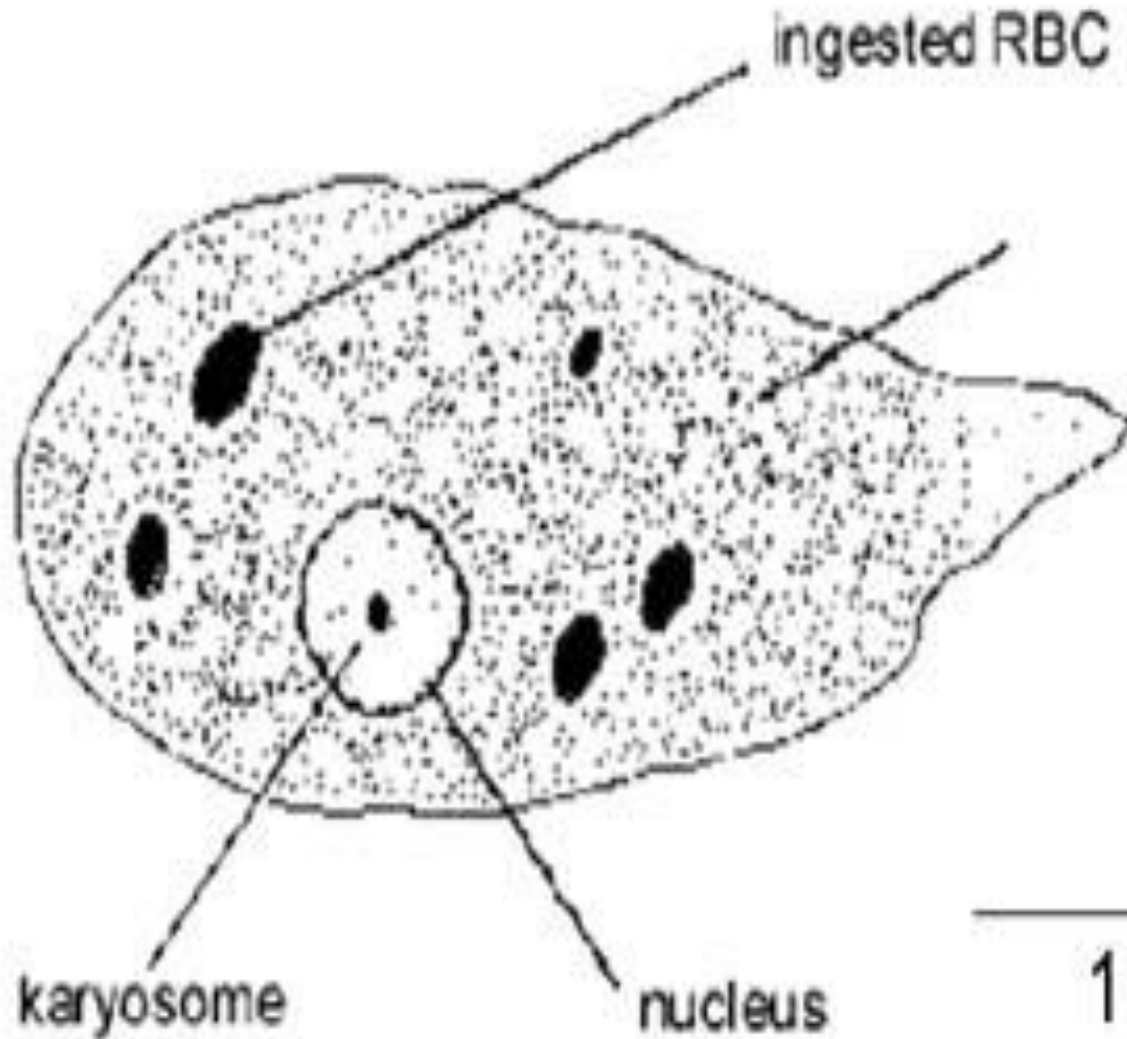
Moves using **pseudopodia** and ingests **RBCs**, a key diagnostic feature.

2-Cyst (infective form):

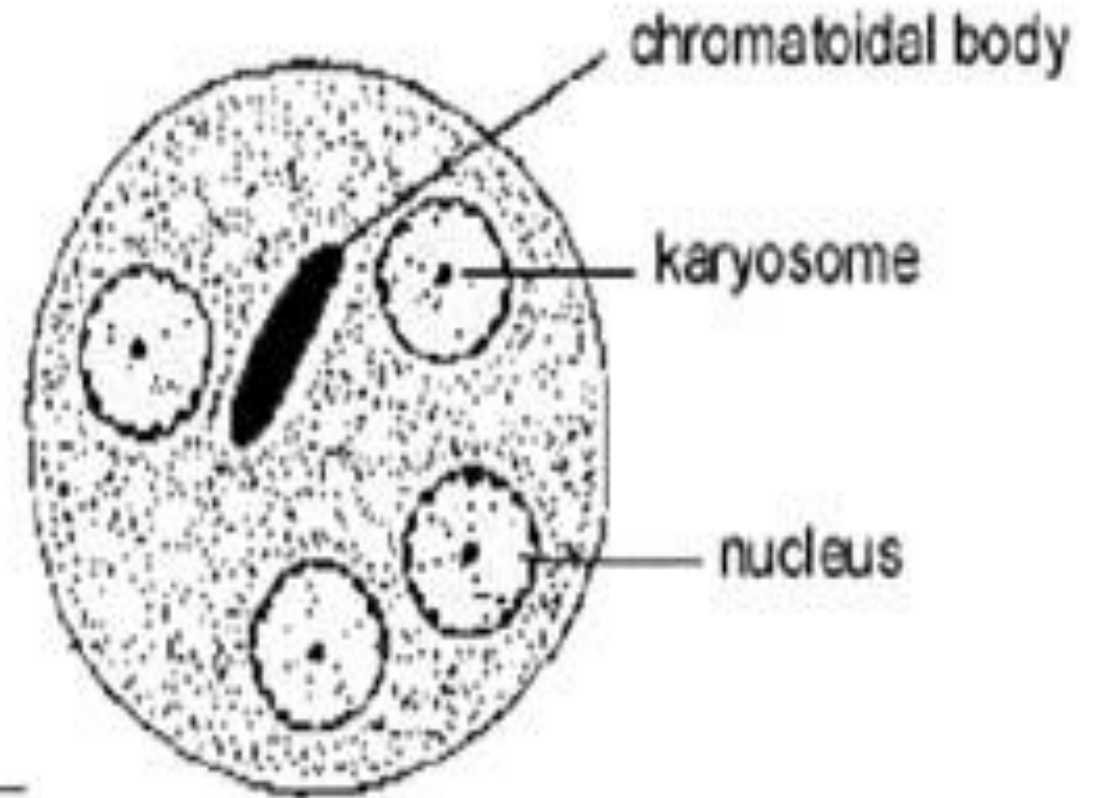
Round, **four nuclei**, with a protective wall.

Can survive outside the host in harsh conditions.

Trophozoite



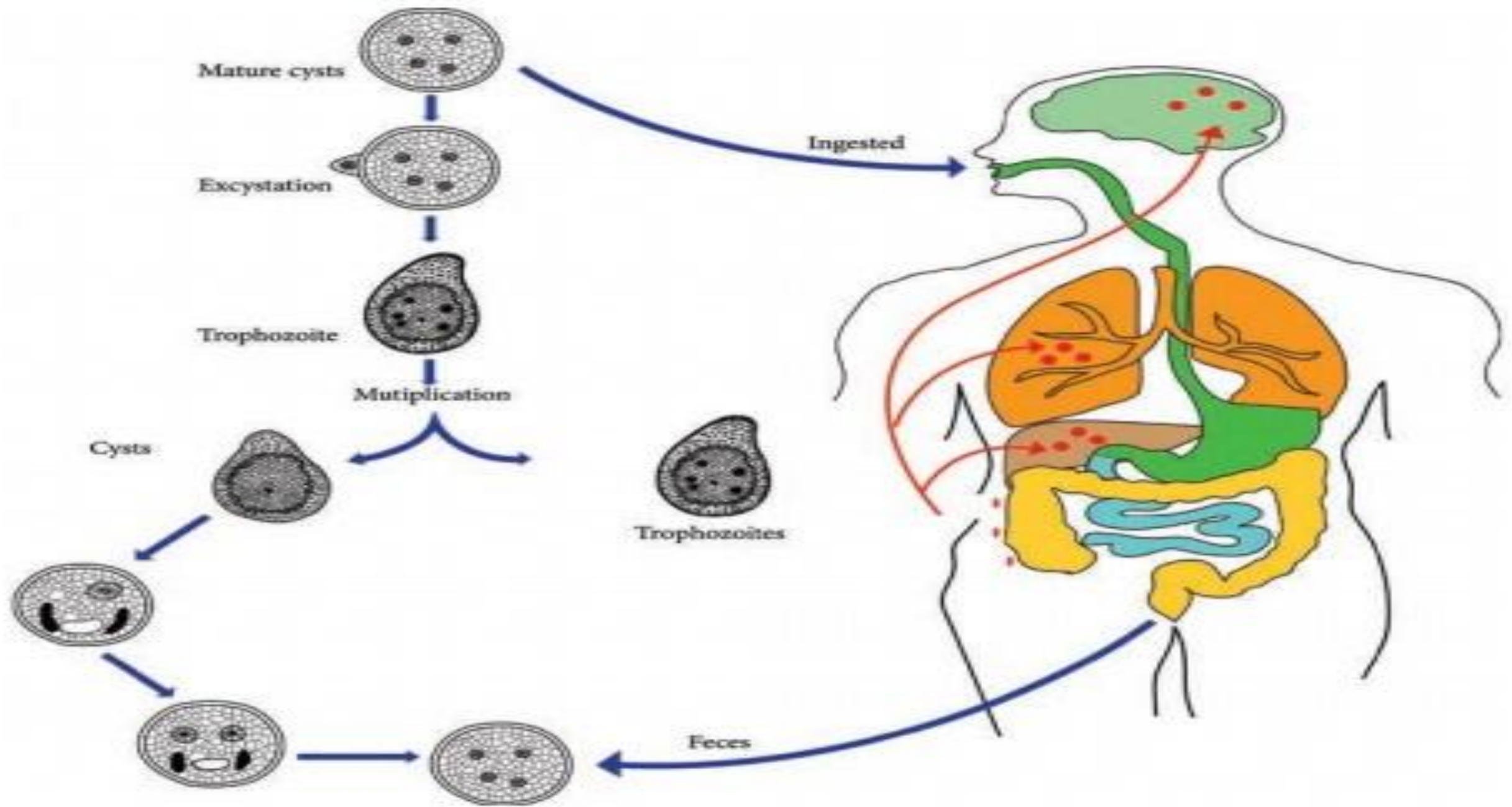
Cyst



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Life Cycle of *Entamoeba histolytica*:

- 1- Ingestion of **cysts** from contaminated food or water.
- 2- Cysts transform into **trophozoites** in the intestine.
- 3- Trophozoites **invade the intestinal mucosa**, causing ulcers.
- 4- Can **spread to the liver** and form abscesses.
- 5- Some trophozoites form **cysts**, which are excreted in feces.



Transmission:

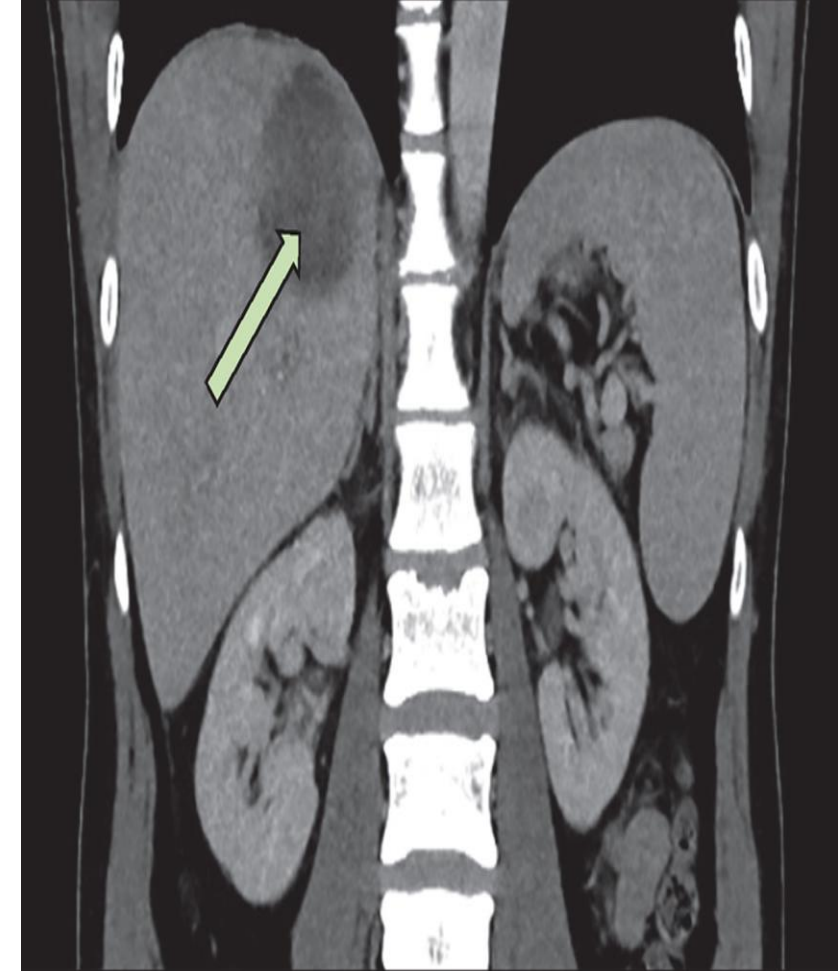
- 1-Fecal-oral route (contaminated food, water, or hands).
- 2-Person-to-person contact (poor hygiene).
- 3-Sexual transmission.

Pathogenesis :

- 1- **Invades colonic mucosa**, leading to ulcer formation (flask-shaped ulcers).
- 2- Can spread via the **portal circulation to the liver**, causing liver abscesses.

Clinical Features

- 1- Dysentery (bloody diarrhea with mucus).
- 2- Abdominal pain and fever.
- 3- Liver abscess (pain in liver, perforation, hepatomegaly).



Diagnosis:

- 1- Microscopy:** Stool sample showing **trophozoites** with ingested RBCs or **cysts**.
- 2- Stool antigen test (ELISA):** highly specific.
- 3- PCR test:** most sensitive.
- 4- Serology:** useful for diagnosing liver abscesses.
- 5-Imaging(Ultrasound/ CT scan):** detects liver abscesses.

Treatment

- 1- **Metronidazole** or **Tinidazole** (kills trophozoites).
- 2- **Paromomycin** or **Iodoquinol** (eliminates cysts from intestines).
- 3- **Drainage of abscesses** if necessary.

Prevention

- 1- Drinking **clean, filtered, or boiled water**.
- 2- Practicing **good hygiene and hand washing**.
- 3- Proper **sanitation**.
- 4- Treating a symptomatic carriers to prevent spread.

Feature	Giardia lamblia	Entamoeba histolytica
1-Transmission	Fecal-oral (contaminated water)	Fecal-oral (contaminated food/water)
2- Infective Form	Cyst	Cyst
3- Trophozoite	Pear-shaped, flagellated	Amoeboid, can ingest RBCs
4- Site of Infection	Small intestine	Large intestine, liver
5- Symptoms	Watery diarrhea, malabsorption	Bloody diarrhea, liver abscess
6- Complications	Chronic diarrhea, weight loss	Liver abscess, perforation
7- Diagnosis	Stool test, antigen test	Stool test, PCR, liver imaging
8- Treatment	Metronidazole, Tinidazole	Metronidazole, Paromomycin



Any Question?



Thank you